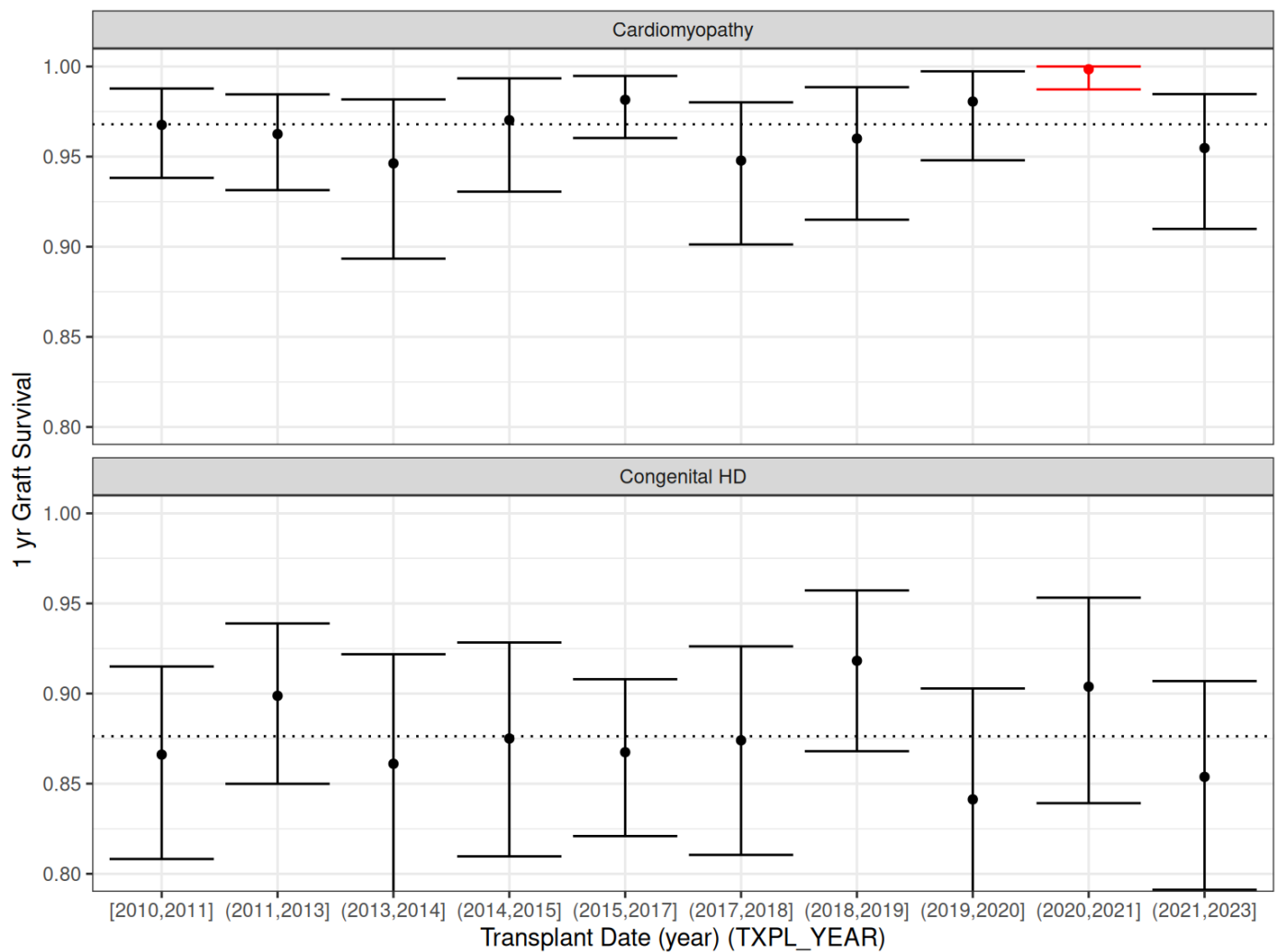




## 6 Simple Calculator

Only considering CHD and Cardiomyopathy patients. This is a basic estimate based on Diagnosis, Age Under 2, and MCS type. Need to make final adjustments for censoring and to prevent overfitting, so consider this a rough estimate.

Cardiomyopathy patients have a predicted survival between 75% (Under 2 years old, on ECMO) and 100% (not on ECMO or VAD).

CHD patients had a predicted survival between 62% (under 2 years old, on ECMO) and 91% (not on ECMO or VAD).

The “-” symbol indicates the value was missing.

### 6.1 Combines CHD and CM

| PRIM_DX       | TXECMO | TXVAD | AGE_UNDER_2 | n_obs | n_survive | p_survive | n_total | n_cens |
|---------------|--------|-------|-------------|-------|-----------|-----------|---------|--------|
| Congenital HD | 1      | 0     | 1           | 34    | 21        | 0.62      | 34      | 0      |

| PRIM_DX        | TXECMO | TXVAD | AGE_UNDER_2 | n_obs | n_survive | p_survive | n_total | n_cens |
|----------------|--------|-------|-------------|-------|-----------|-----------|---------|--------|
| Cardiomyopathy | 1      | 0     | 1           | 8     | 6         | 0.75      | 10      | 2      |
| Congenital HD  | 1      | 0     | 0           | 16    | 12        | 0.75      | 19      | 3      |
| Congenital HD  | 0      | 1     | 1           | 105   | 83        | 0.79      | 125     | 20     |
| Congenital HD  | -      | -     | 0           | 18    | 15        | 0.83      | 19      | 1      |
| Congenital HD  | 0      | 1     | 0           | 100   | 87        | 0.87      | 107     | 7      |
| Congenital HD  | 0      | 0     | 1           | 445   | 397       | 0.89      | 473     | 28     |
| Cardiomyopathy | 0      | 1     | 1           | 157   | 141       | 0.90      | 175     | 18     |
| Congenital HD  | 0      | 0     | 0           | 613   | 551       | 0.90      | 675     | 62     |
| Congenital HD  | -      | -     | 1           | 11    | 10        | 0.91      | 11      | 0      |
| Cardiomyopathy | 1      | 0     | 0           | 25    | 23        | 0.92      | 25      | 0      |
| Cardiomyopathy | 0      | 1     | 0           | 426   | 417       | 0.98      | 480     | 54     |
| Cardiomyopathy | 0      | 0     | 0           | 479   | 469       | 0.98      | 528     | 49     |
| Cardiomyopathy | 0      | 0     | 1           | 201   | 198       | 0.99      | 211     | 10     |
| Cardiomyopathy | -      | -     | 1           | 5     | 5         | 1.00      | 6       | 1      |
| Cardiomyopathy | -      | -     | 0           | 8     | 8         | 1.00      | 9       | 1      |

## 6.2 CM Only

| PRIM_DX        | TXECMO | TXVAD | AGE_UNDER_2 | n_obs | n_survive | p_survive | n_total | n_cens |
|----------------|--------|-------|-------------|-------|-----------|-----------|---------|--------|
| Cardiomyopathy | 1      | 0     | 1           | 8     | 6         | 0.75      | 10      | 2      |
| Cardiomyopathy | 0      | 1     | 1           | 157   | 141       | 0.90      | 175     | 18     |
| Cardiomyopathy | 1      | 0     | 0           | 25    | 23        | 0.92      | 25      | 0      |
| Cardiomyopathy | 0      | 1     | 0           | 426   | 417       | 0.98      | 480     | 54     |
| Cardiomyopathy | 0      | 0     | 0           | 479   | 469       | 0.98      | 528     | 49     |
| Cardiomyopathy | 0      | 0     | 1           | 201   | 198       | 0.99      | 211     | 10     |
| Cardiomyopathy | -      | -     | 1           | 5     | 5         | 1.00      | 6       | 1      |
| Cardiomyopathy | -      | -     | 0           | 8     | 8         | 1.00      | 9       | 1      |

## 6.3 CHD Only

Adding in Laterality Disorder sub-category. With the help of ChatGPT (needs review!) I used these CHD conditions to define a new Laterality Disorder ( **CHD\_LAT** ) sub-diagnosis:

- Dextrocardia (CHD\_DEX)
- Situs Inversus (CHD\_SI)

- Heterotaxy (CHD\_HETER)
- Interrupted IVC (CHD\_IIVC)
- Bilateral SVC (CHD\_BIVC)
- Left SVC with no right SVC (CHD\_L SVC)
- Right Aortic Arch (CHD\_RAA)
- AV Discordance when part of atrial or ventricular situs abnormality (CHD\_AVD)

The first three (Dextrocardia, Heterotaxy, and Situs Inversus) were significant in the univariate analysis.

| PRIM_DX       | CHD_LAT | TXECMO | TXVAD | AGE_UNDER_2 | n_obs | n_survive | p_survive | n_total | n_cens |
|---------------|---------|--------|-------|-------------|-------|-----------|-----------|---------|--------|
| Congenital HD | 1       | 1      | 0     | 1           | 1     | 0         | 0.00      | 1       | 0      |
| Congenital HD | 0       | 1      | 0     | 1           | 33    | 21        | 0.64      | 33      | 0      |
| Congenital HD | 1       | 0      | 1     | 1           | 6     | 4         | 0.67      | 7       | 1      |
| Congenital HD | 1       | 0      | 0     | 1           | 23    | 17        | 0.74      | 24      | 1      |
| Congenital HD | 0       | 1      | 0     | 0           | 16    | 12        | 0.75      | 19      | 3      |
| Congenital HD | 0       | 0      | 1     | 1           | 99    | 79        | 0.80      | 118     | 19     |
| Congenital HD | 1       | 0      | 0     | 0           | 42    | 34        | 0.81      | 52      | 10     |
| Congenital HD | 0       | -      | -     | 0           | 18    | 15        | 0.83      | 18      | 0      |
| Congenital HD | 1       | 0      | 1     | 0           | 6     | 5         | 0.83      | 8       | 2      |
| Congenital HD | 0       | 0      | 1     | 0           | 94    | 82        | 0.87      | 99      | 5      |
| Congenital HD | 0       | 0      | 0     | 1           | 422   | 380       | 0.90      | 449     | 27     |
| Congenital HD | 0       | 0      | 0     | 0           | 571   | 517       | 0.91      | 623     | 52     |
| Congenital HD | 0       | -      | -     | 1           | 11    | 10        | 0.91      | 11      | 0      |
| Congenital HD | 1       | -      | -     | 0           | 0     | 0         | -         | 1       | 1      |

## 7 Variable Descriptions

Show 

10

 entries

Search:

|   | var            | descr                          | primary           | secondary      |
|---|----------------|--------------------------------|-------------------|----------------|
|   | <div>All</div> | <div>All</div>                 | <div>All</div>    | <div>All</div> |
| 1 | LSABOINC       | ABO Incompatible at Listing    | ABO Incompatible  |                |
| 2 | TXABOINC       | ABO Incompatible at Transplant | ABO Incompatible  |                |
| 3 | PTLT18         | Age 18 at listing              | Age 18 at listing |                |
| 4 | CHD_AA         | CHD: Aortic Atresia            | CHD               | Aortic Atresia |

|    | var        | descr                                        | primary | secondary                               |
|----|------------|----------------------------------------------|---------|-----------------------------------------|
| 5  | CHD_AHIH   | CHD: Arch Hypoplasia/Interruption/Hypoplasia | CHD     | Arch Hypoplasia/Interruption/Hypoplasia |
| 6  | CHD_ALCAPA | CHD: ALCAPA                                  | CHD     | ALCAPA                                  |
| 7  | CHD_ANOM   | CHD: Coronary Anomaly                        | CHD     | Coronary Anomaly                        |
| 8  | CHD_AR     | CHD: Aoritc Regurgitation                    | CHD     | Aoritc Regurgitation                    |
| 9  | CHD_ASPL   | CHD: Heterotaxy: Asplenia                    | CHD     | Heterotaxy: Asplenia                    |
| 10 | CHD_AVD    | CHD: AV Discordance                          | CHD     | AV Discordance                          |

Showing 1 to 10 of 486 entries